Bhavdeep Arora

416-841-5785 | bhavdeepsa@gmail.com | linkedin/in/bhavdeeparora | github.com/bhavv04 | bhavdeeparora.dev

Education

Toronto Metropolitan University — Toronto, ON
Bachelor of Science in Computer Science, Concentration in Software Engineering — *October 2026*

Experience

Volunteer Data Engineer — Oct. 2026 – Feb. 2027 (Expected)
Develop for Good — *Remote*

- Prepare and organize nonprofit partner data using Python-based ingestion and transformation workflows to support operational analysis.
- Apply data-cleaning and validation checks to improve consistency, resolve quality issues, and prepare reliable datasets for reporting.
- Translate analytical findings into concise reports and recommendations for non-technical stakeholders.

Volunteer Data Analyst — May 2026 – Aug. 2026
Data for Good Toronto — *Toronto, ON*

- Built a Python data pipeline to collect and process ETF data for a nonprofit partner, automating a previously manual research workflow.
- Cleaned, normalized, and structured data from inconsistent sources to produce reliable datasets for downstream analysis.
- Documented pipeline outputs and communicated findings through a report supporting partner research and decision-making.

Data Analyst (Contract) — May 2025 – Apr. 2026
Toronto Police Service — *Toronto, ON*

- Used SQL, Python, and R to analyze operational datasets, investigate trends, and answer business and stakeholder questions.
- Developed Power BI reports and dashboards to communicate operational metrics and analytical findings to non-technical audiences.
- Validated datasets, investigated inconsistencies, and translated analytical results into actionable insights for business analysis.

Projects

GAIA | *Python, Polars, PyTorch Geometric, NetworkX, MLflow* — Aug. 2026 – Present

- Designed a multi-source data pipeline integrating NOAA, NASA Earthdata, OpenAQ, iNaturalist, and SoilGrids environmental data into a unified schema.
- Processed heterogeneous time-series and geospatial datasets with Polars, applying cleaning, alignment, and feature engineering to prepare model-ready data.
- Developed an MLflow-tracked graph neural network to predict ecological cascade failures, achieving 0.84 F1 versus 0.68 for a logistic regression baseline.

Deadzones | *Python, Pandas, NumPy, Scikit-learn* — May 2026 – Aug. 2026

- Integrated NOAA/LUMCON dead-zone surveys, USGS nitrogen-loading data, and the NOAA Oceanic Niño Index into a validated environmental dataset spanning 1985–2024.
- Built reproducible preprocessing and feature-engineering workflows for environmental time series, including lagged variables, rolling means, and interaction features.
- Evaluated model configurations using leave-one-out cross-validation, paired Wilcoxon tests, bootstrap confidence intervals, and permutation importance.

Funes | *Rust, SQL, Ollama, TypeScript, React* | GitHub — Apr. 2026 – Jul. 2026

- Built a local-first ingestion and retrieval pipeline that indexes 5,000+ files, notes, and shell-history items for semantic search using vector embeddings.
- Measured 12 ms average query latency and 94% accuracy against a ground-truth query set.

Technical Skills

Languages: Python, SQL, R, Rust, Go, Java, JavaScript, TypeScript
Data Engineering: ETL/ELT, Data Ingestion, Data Cleaning, Transformation, Data Validation, Feature Engineering
Data Tools: Pandas, Polars, NumPy, MySQL, SQLite, MongoDB, Apache Spark, Hadoop, MLflow, Power BI
Cloud & DevOps: AWS, Azure, Google Cloud Platform, Docker, Git, GitHub Actions, CI/CD